

Multiple group item response theory in LEVANTE

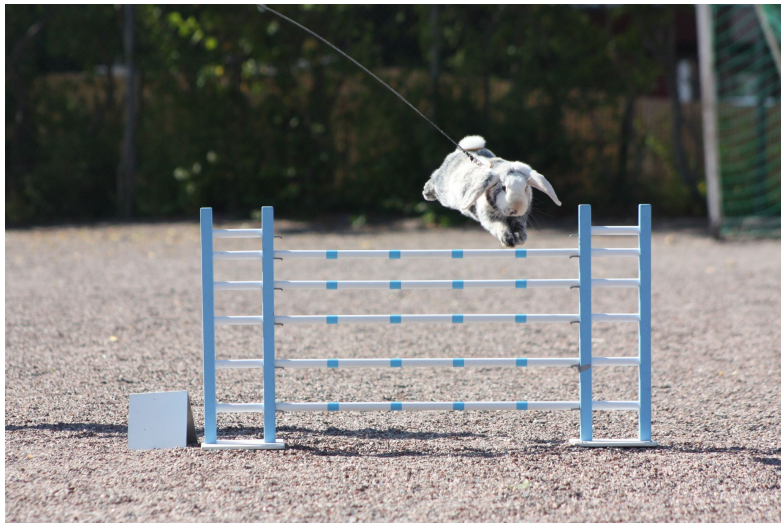
LEVANTE Hackathon 2026

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Measurement in psychology is hard

How good is a rabbit at jumping?



https://commons.wikimedia.org/wiki/File:Rabbit_jumping_over_a_fence_at_Rabbit_Show_Jumping.jpg

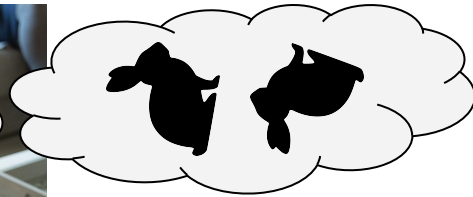
 Fruitpunchli

How good is a human at spatial reasoning?

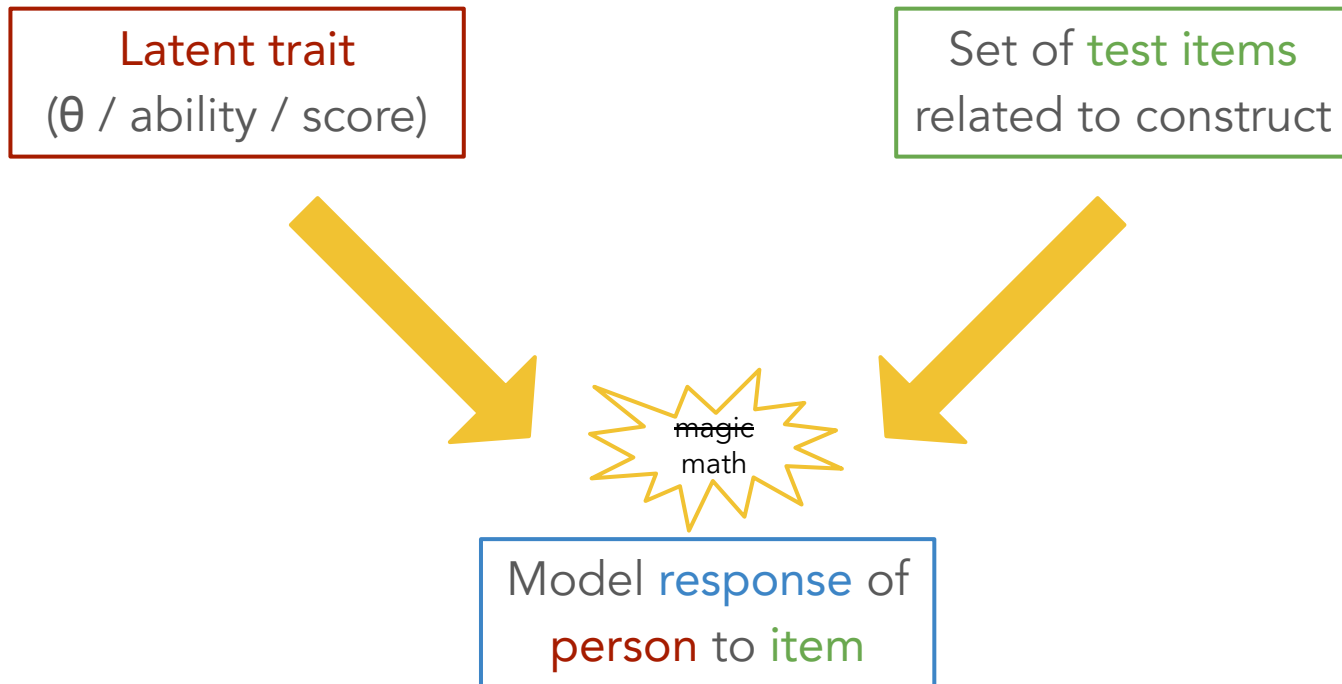


https://commons.wikimedia.org/wiki/File:A_young_child_engages_with_colorful_toys_on_a_table_in_a_bright_living_room.jpg

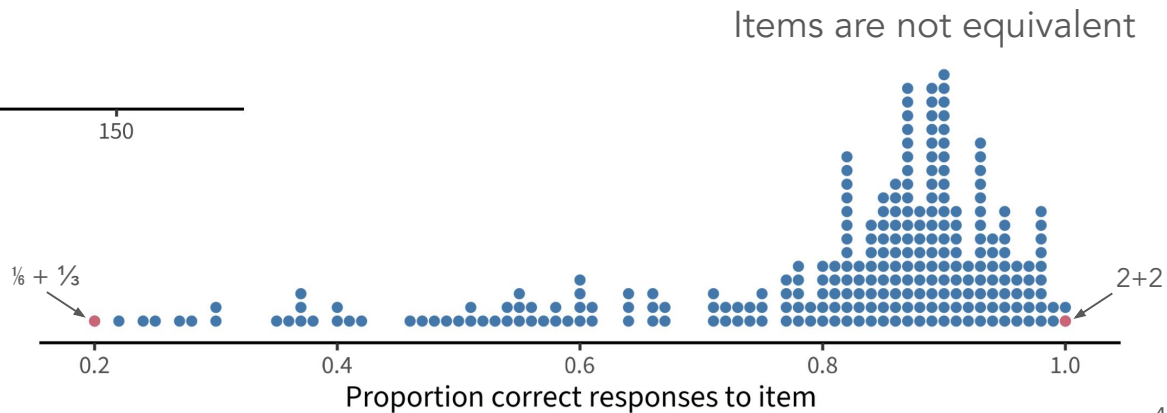
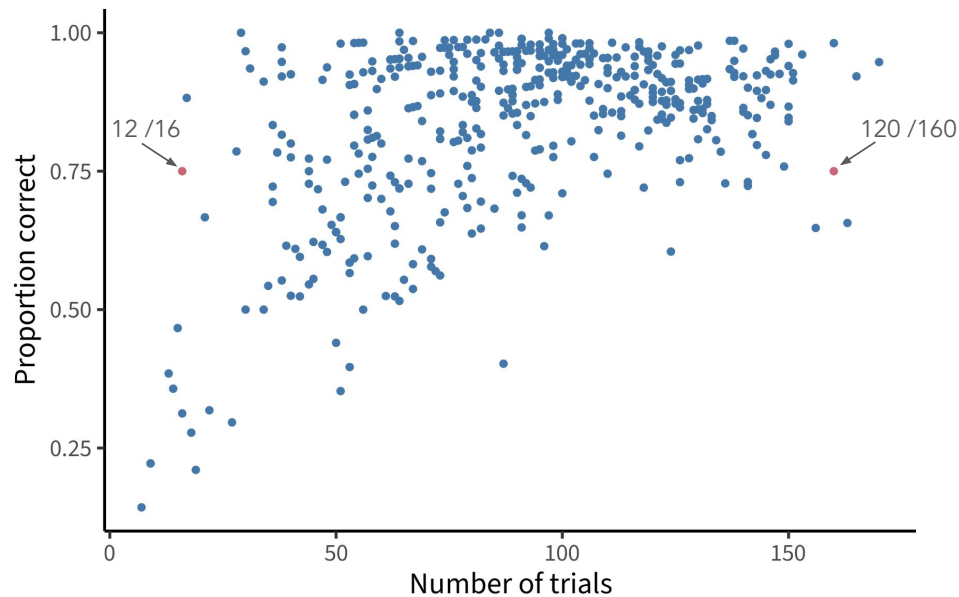
 Shixart1985



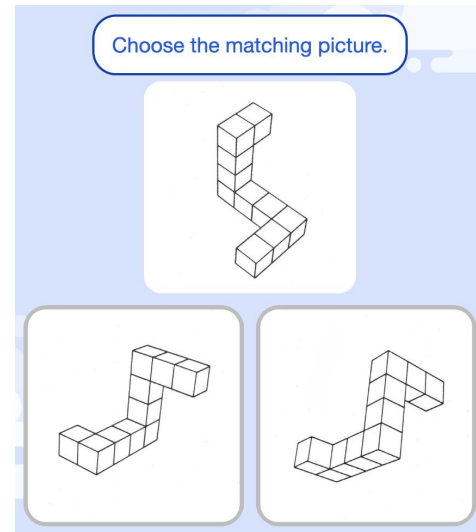
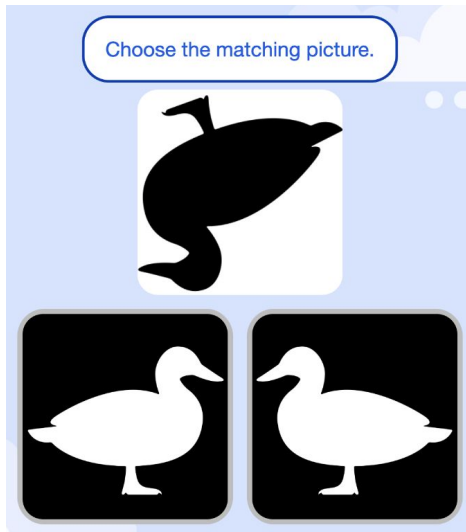
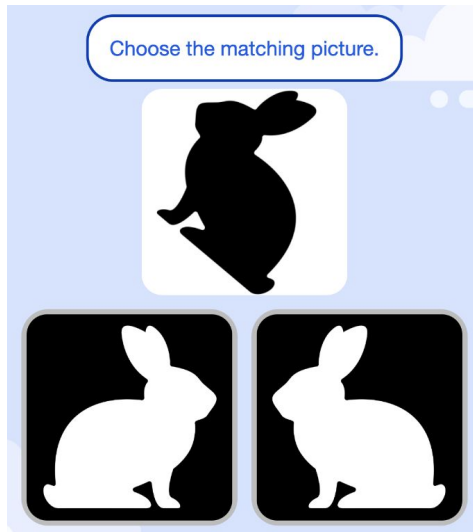
IRT building blocks

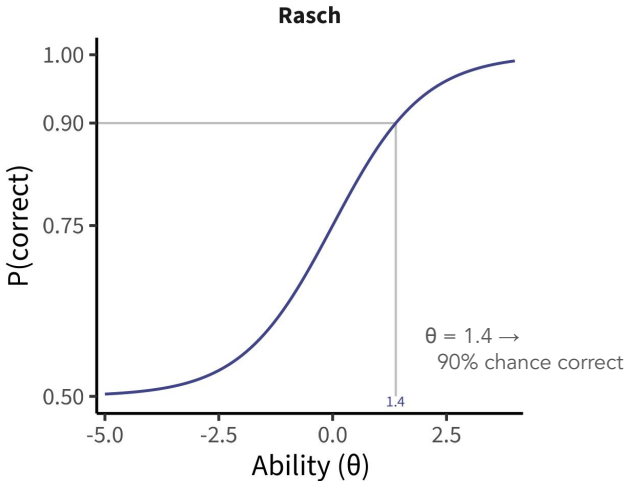
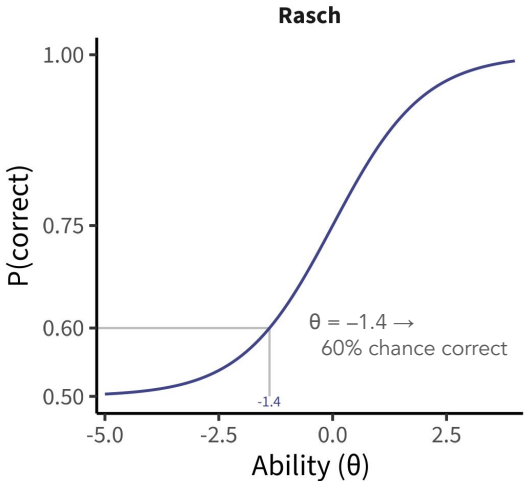
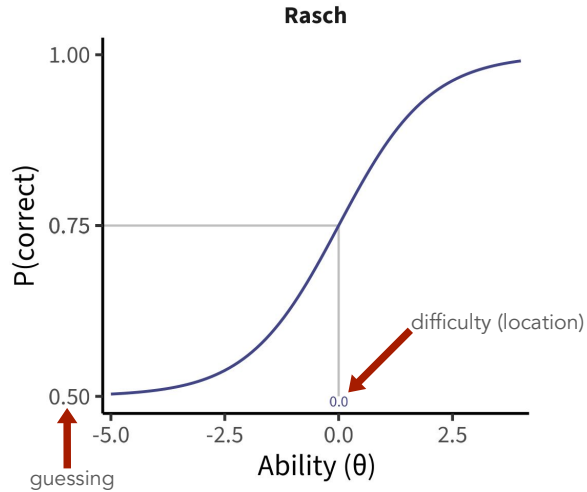


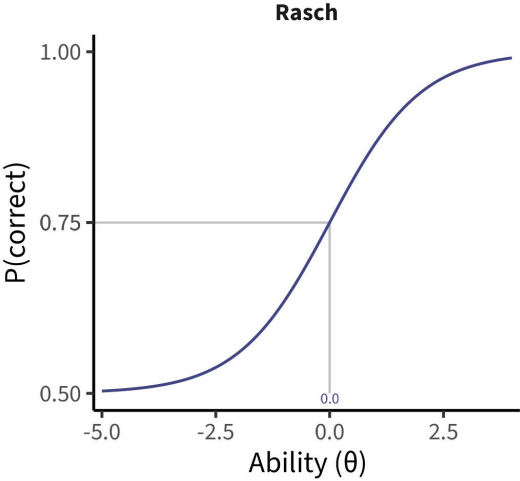
Accuracy is not enough



Shape rotation task

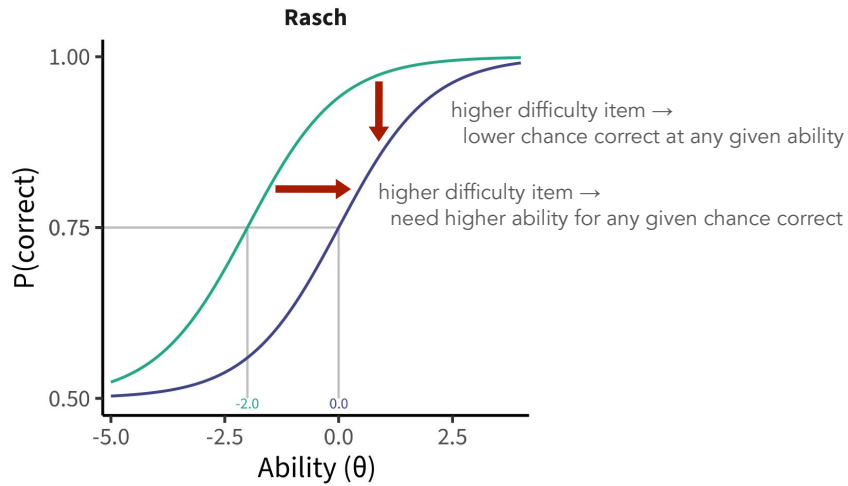






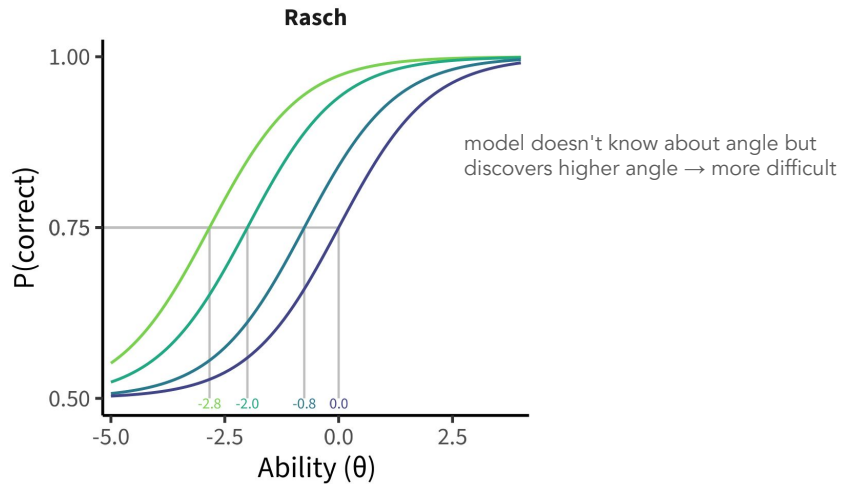
Angle — 160





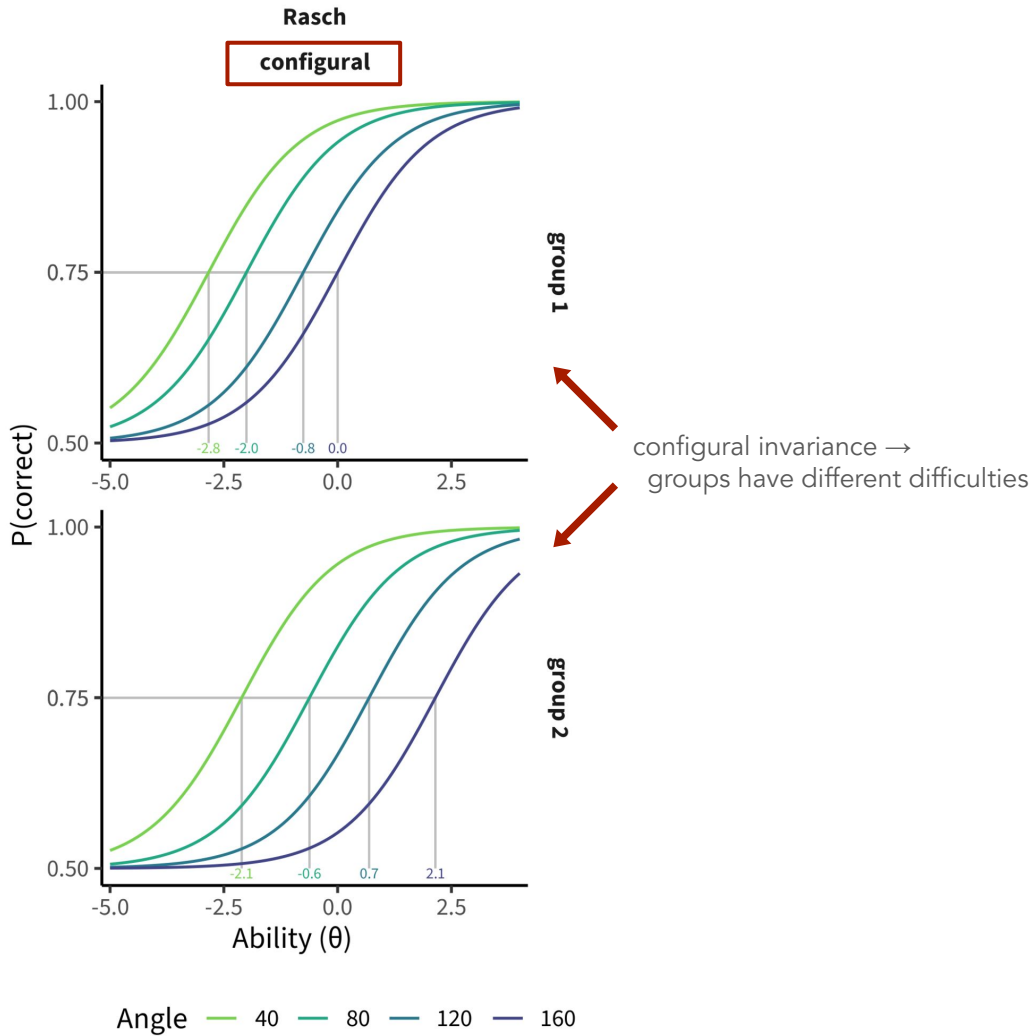
Angle — 80 — 160



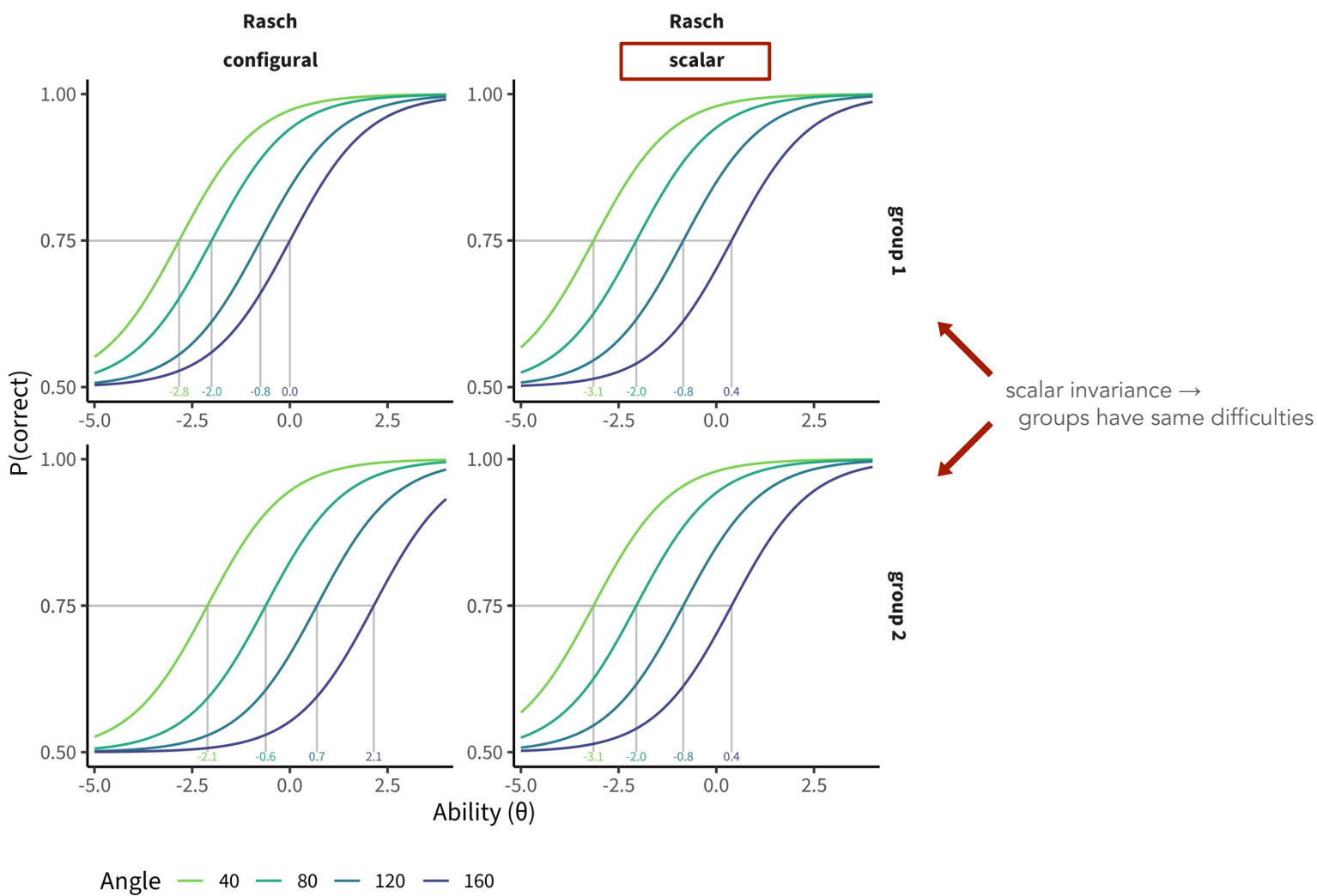


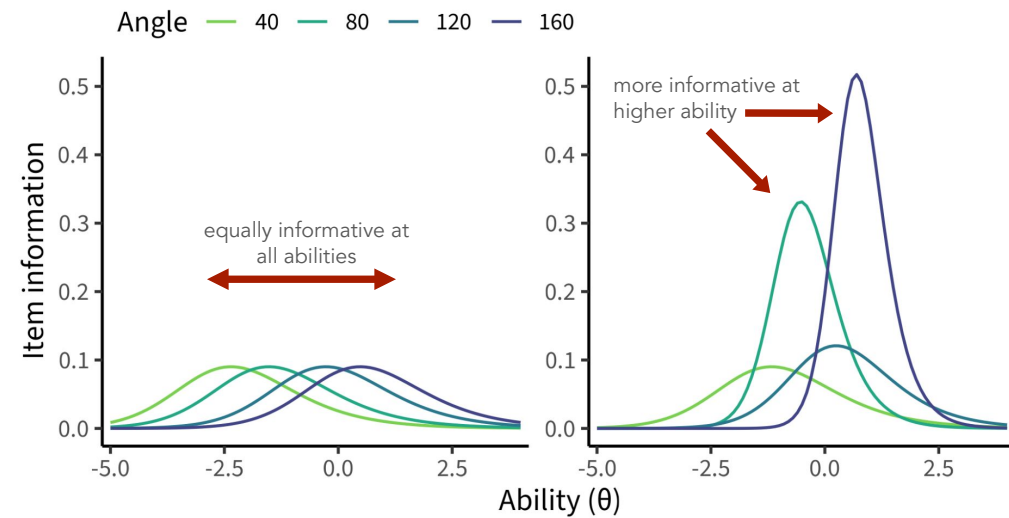
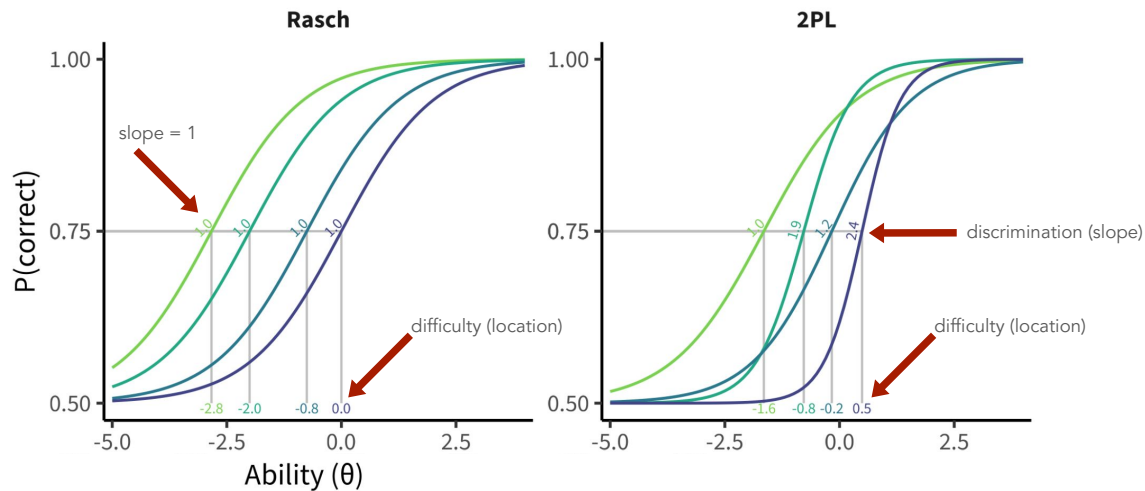
Angle — 40 — 80 — 120 — 160



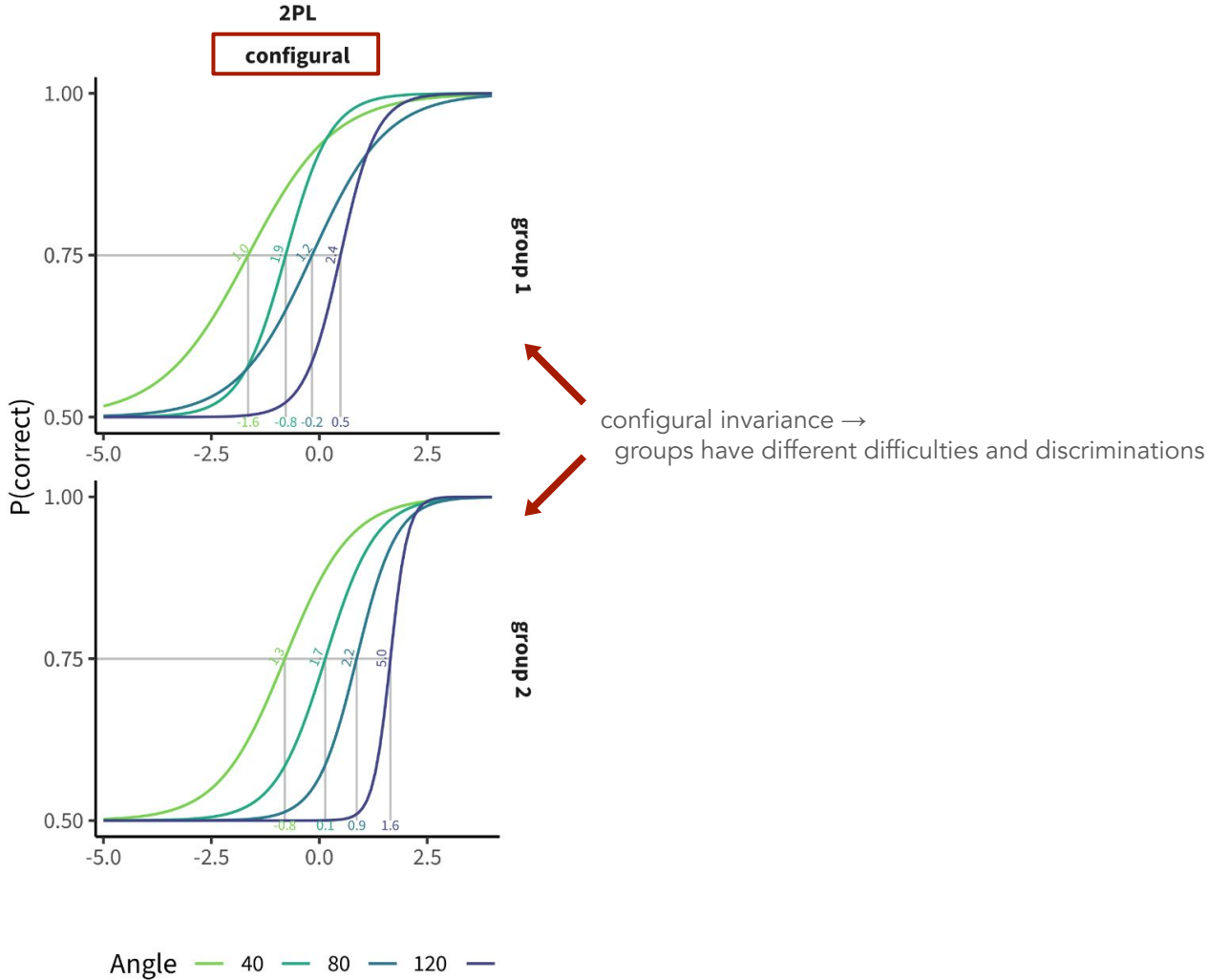


Rasch
invariance





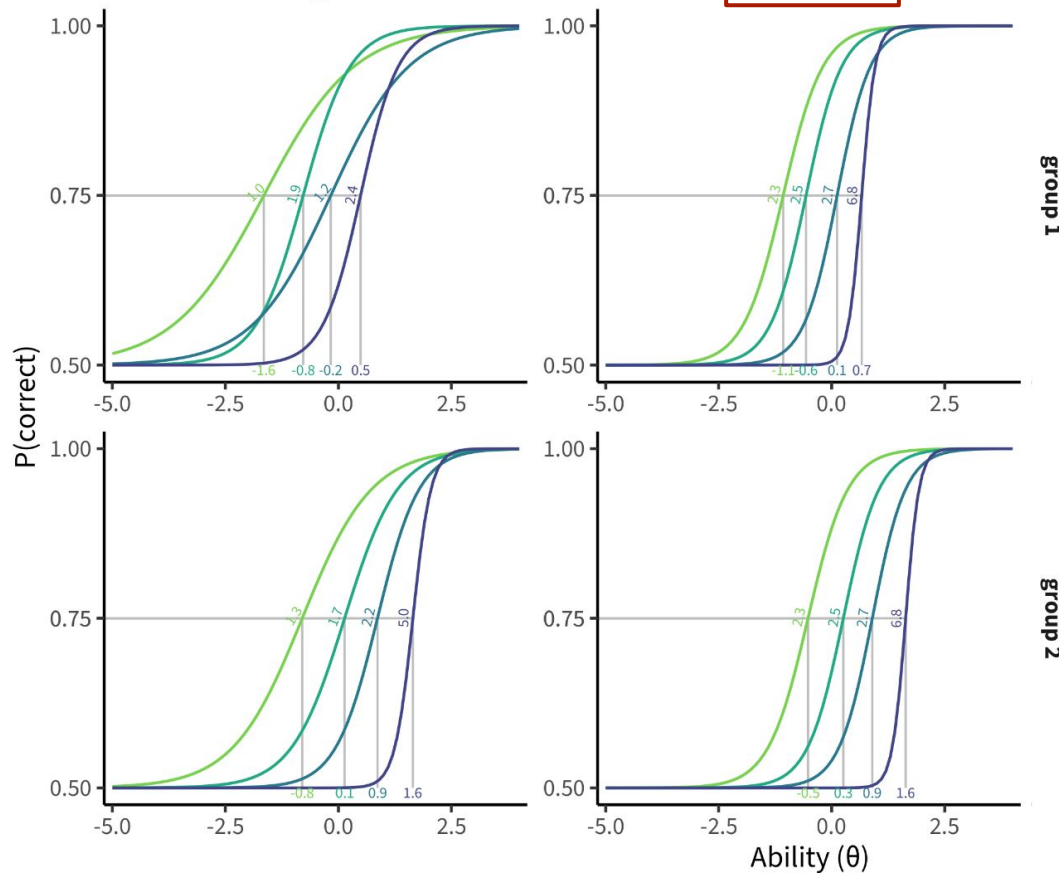
2PL
invariance



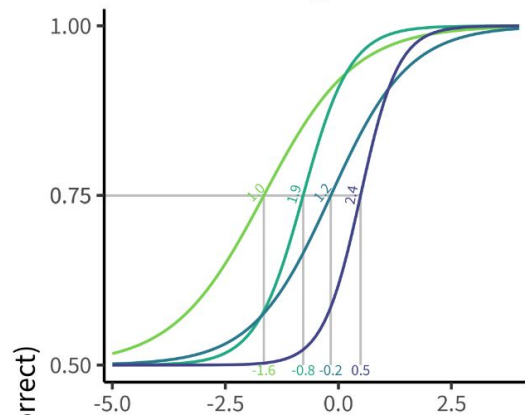
2PL
invariance

2PL
configural

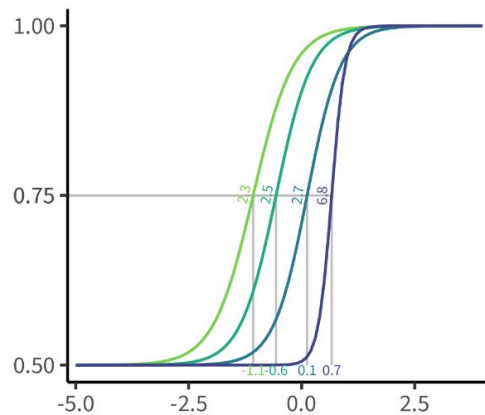
2PL
metric



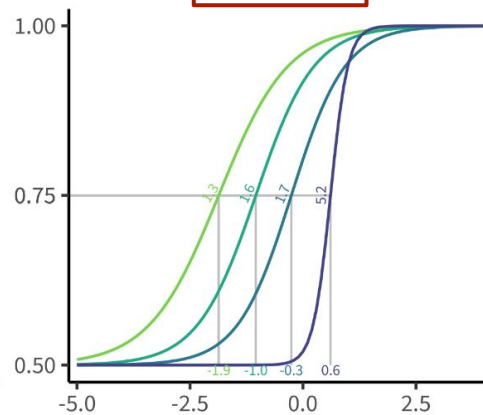
2PL
configural



2PL
metric



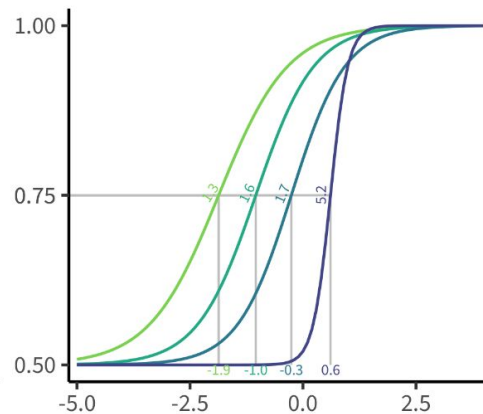
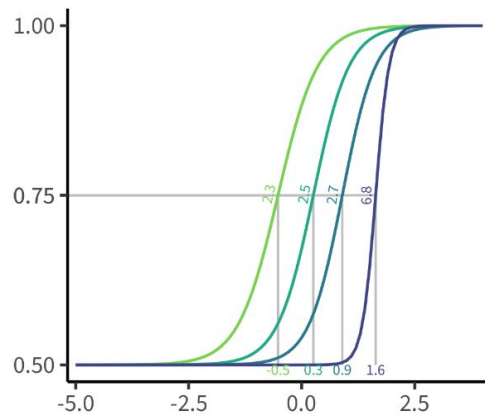
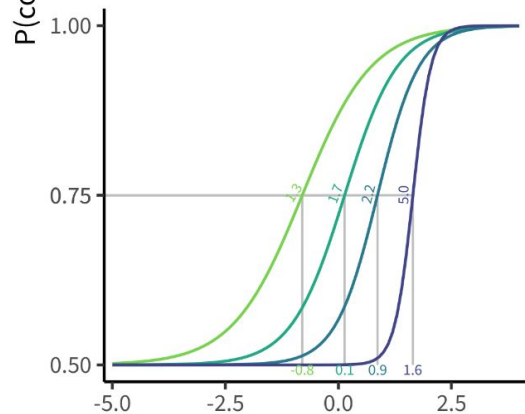
2PL
scalar



2PL
invariance

group 1

scalar invariance $\rightarrow$
groups have same
difficulties and
discriminations



group 2

Ability (θ)

Angle — 40 — 80 — 120 — 160

Model comparison

Bayesian Information Criterion (BIC)

how well does model fit data, without overfitting

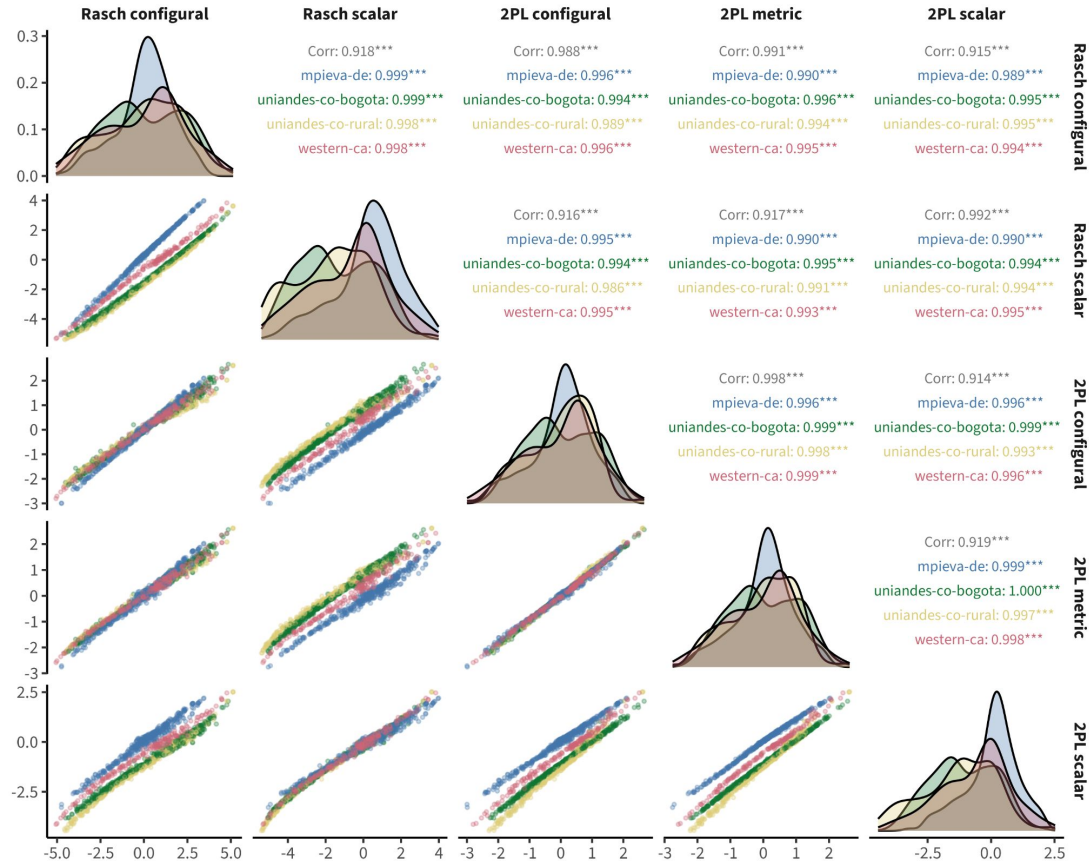
Model	Invariance	BIC
Rasch	configural	53643
Rasch	scalar	53550
2PL	configural	53873
2PL	metric	53484
2PL	scalar	53330

Marginal reliability

how much variance in scores is captured by scale

Site	Rasch	2PL
mpieva-de	0.56	0.90
uniandes-co-bogota	0.53	0.83
uniandes-co-rural	0.50	0.78
western-ca	0.51	0.81

Comparing scores



Measurement invariance

Interpretation

Consequence

No invariance

Fractions are part of math in group 1, part of sentence understanding in group 2

Scores don't measure same construct across sites

Configural invariance

Fractions are easier in group 1 for a similar ability child than they are in group 2

Scores measure math ability within site but can't be compared across sites

Scalar invariance

Fractions are equally easy for a child of the same ability in group 1 and group 2

Scores are on the same scale and can be compared across sites

mirt: fitting models

IRT

```
mirt(  
  data = df,  
  itemtype = "Rasch",  
  guess = 0.5,  
  model = 1  
)
```

columns = items
rows = observations
values = integers

"Rasch", "2PL", etc

single value or value per item

1 = unidimensional
can be string with constraints, priors, etc

Multigroup IRT

```
multipleGroup(  
  data = df,  
  itemtype = "Rasch",  
  guess = 0.5,  
  model = 1,  
  group = df_groups,  
  invariance = ""  
)
```

same as above

string for each row

configural → ""
metric → c("slopes")
scalar → c("free_means", "free_var", "intercepts", "slopes")

mirt: extracting from models

parameters

```
coef(mod, simplify = TRUE)
```

→ easiness

```
coef(mod, simplify = TRUE,  
      IRTpars = TRUE)
```

→ difficulty



scores

```
fscores(mod, full.scores.SE = TRUE)
```

various elements – see ?extract.mirt

```
extract.mirt(mod, "BIC")
```

marginal reliability (mirt)

```
marginal_rxx(mod)
```

marginal reliability (multipleGroup)

```
mod_g1 <- extract.group(mod, "group1")  
marginal_rxx(mod_g1)
```

Complications

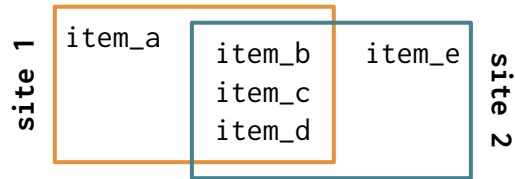
Problem: participants can see given item multiple times

Solution: de-duplicate items into instances and constrain instances to have equal parameters

id	item_uid	item_inst
1	item_a	item_a-1
1	item_a	item_a-2
1	item_b	item_b-1
1	item_b	item_b-2

Problem: items differ across sites

Solution: first fit model to overlapping items, then fit model to all items with parameters for overlapping items fixed to their value from first model



Thank you!

slides

mikabr.io/levante-multigroup

tutorial

github.com/levante-hackathon-2026/public_curriculum/tree/main/irt-in-levante

R package

levante-framework.github.io/levantemodels

other resources

levante-hackathon-2026.github.io/resources

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Office hours

Fri Jul 10 10:30 PDT